

Paper Overview: Methods, Models and Alternatives

Trump Posts vs. S&P 500

1) Preprocessing

Applied approach	What is this?	Why it fits here	Advantages / Disadvantages	Relevant alternatives and why not
NLTK-based preprocessing with tokenization, POS-tagging, lemmatization, and stopword removal	A standard preprocessing pipeline that cleans and normalizes text while preserving linguistically meaningful forms.	It suits a research-oriented NLP pipeline where preprocessing steps should remain transparent and interpretable.	Advantages: flexible, widely used in research, supports POS-aware lemmatization. Disadvantages: slower and less production-oriented than spaCy.	spaCy is faster and more efficient, but NLTK is sufficient here and better aligned with a transparent research workflow.
Lemmatization instead of stemming	Reduces words to valid base forms using linguistic information and POS context.	It improves interpretability and produces cleaner input for downstream analysis.	Advantages: more accurate than stemming, returns valid dictionary forms. Disadvantages: slower and more computationally expensive.	Stemming is faster, but too crude because it may create non-real roots and reduce interpretability.

2) Sentiment Analysis

Applied approach	What is this?	Why it fits here	Advantages / Disadvantages	Relevant alternatives and why not
VADER	A rule-based sentiment model designed for social-media language and sensitive to punctuation, capitalization, and emphasis.	Trump posts are short, rhetorical, and stylistically marked, which makes VADER a strong fit.	Advantages: no training data needed, transparent, handles capitalization and punctuation well, strong for social-media text. Disadvantages: weaker with sarcasm, deeper context, and domain-specific financial meaning.	AFINN is simpler but less adapted to social media. ML-based sentiment models can capture more nuance, but they require labeled data.
FinBERT	A transformer-based sentiment model trained on financial language.	It fits because many posts concern markets, trade, the economy, and policy-relevant topics.	Advantages: captures financial nuance and contextual meaning better than generic lexicon tools. Disadvantages: more computationally demanding and not specifically trained on political social-media language.	VADER only would miss financial semantics. A custom SVM or neural sentiment classifier would require labeled in-domain data, which the project does not have. FinBERT is therefore a useful domain-specific extension beyond the lecture core.
Combined VADER + FinBERT setup	A dual-model sentiment design that captures both rhetorical and financial tone.	This is useful because stylistic sentiment and financial sentiment are not identical in the posts.	Advantages: more robust, captures complementary sentiment dimensions. Disadvantages: adds complexity and may create disagreement between models.	A single-model sentiment setup would provide a narrower view of the posts and miss either rhetorical or domain-specific sentiment.

3) Topic Modelling

Applied approach	What is this?	Why it fits here	Advantages / Disadvantages	Relevant alternatives and why not
BERTopic	A transformer-based topic modelling approach that combines embeddings, clustering, and class-based TF-IDF .	It is well suited to short political posts where semantic similarity matters more than simple word counts.	Advantages: strong semantic performance, especially on short texts, and more context-sensitive than bag-of-words methods. Disadvantages: computationally expensive and methodologically more complex.	LDA is bag-of-words based and performs poorly on short texts. Top2Vec is closer conceptually, but BERTopic typically offers stronger transformer-based semantic representations and clearer topic extraction for this kind of data.
Manual topic labelling after model output	Human-readable labels are assigned to discovered topics after inspection.	This makes the model output understandable and usable in an academic paper.	Advantages: improves interpretability and reporting quality. Disadvantages: introduces some subjective judgment.	Purely automatic topic naming is often noisier and less suitable for paper presentation.

4) Semantic Representation and Clustering

Applied approach	What is this?	Why it fits here	Advantages / Disadvantages	Relevant alternatives and why not
all-MiniLM-L6-v2 sentence embeddings	A sentence-transformer model that encodes complete posts into dense semantic vectors.	The project works with whole posts and daily semantic patterns, so sentence-level meaning is required.	Advantages: strong sentence-level semantics, compact representation, suitable for clustering and similarity analysis. Disadvantages: less interpretable than lexical features and more computationally demanding than classical embeddings.	Word2Vec produces static word-level embeddings, not contextual sentence representations. It is therefore less suitable for post-level clustering, similarity, and novelty detection.
K-Means on embeddings	A clustering method that groups semantically similar post embeddings.	It fits naturally once posts have been encoded as vectors.	Advantages: simple, scalable, and easy to explain. Disadvantages: requires choosing K and may oversimplify cluster structure.	LDA models co-occurring words rather than semantic distances in embedding space, so it is less aligned with an embedding-first pipeline.
UMAP visualization	A dimensionality-reduction method used to project embeddings into 2D for inspection.	It helps visualize whether semantic groupings are meaningfully separated.	Advantages: intuitive visual interpretation of embedding space. Disadvantages: mainly descriptive and may distort global distances.	Included primarily as a visualization tool, no need for alternatives.

5) Novelty and Feature Engineering

Applied approach	What is this?	Why it fits here	Advantages / Disadvantages	Relevant alternatives and why not
Cosine-distance novelty score	A custom feature measuring how unusual a day's semantic content is relative to recent posting history.	It fits the research question because unusual semantic shifts may matter more to markets than routine rhetoric.	Advantages: captures semantic drift beyond average sentiment or topic flags, adds an original feature to the study. Disadvantages: less standard and somewhat harder to interpret.	There is no direct lecture equivalent. This is a project-specific feature engineering step extending the lecture material .
Daily aggregation of post-level features	Aggregates post-level outputs into day-level variables aligned with market outcomes.	Necessary because the prediction target is daily S&P 500 movement.	Advantages: creates an interpretable feature matrix aligned with financial data. Disadvantages: loses within-day sequence information.	Sequence-based neural approaches would preserve more temporal detail, but they are much more complex and do not fit the chosen daily prediction design as well.

6) Prediction and Modelling

Applied approach	What is this?	Why it fits here	Advantages / Disadvantages	Relevant alternatives and why not
Next-day classification and regression setup	A predictive framework using day t text features for day $t+1$ market outcomes.	It directly matches the research question while preserving temporal order and avoiding look-ahead bias.	Advantages: methodologically clean and easy to interpret. Disadvantages: broad market prediction remains noisy.	A purely descriptive or event-study-only design would be simpler, but it would test predictive signal less directly.
Logistic Regression, Linear SVM, Random Forest, Gradient Boosting	A benchmark set of linear and non-linear classifiers.	This allows comparison across interpretable and more flexible models.	Advantages: strong comparative framework and robust benchmarking. Disadvantages: increases model complexity and reporting effort.	Naive Bayes is a useful lecture baseline, but its independence assumption is restrictive for richer engineered NLP features. Logistic Regression remains important and is already included as an interpretable benchmark.
ElasticNet and Random Forest Regressor	Linear-regularized and non-linear regression models for return magnitude prediction.	They provide a reasonable contrast between regularized linear modelling and flexible tree-based modelling.	Advantages: cover different functional forms and handle correlated features better than simple OLS. Disadvantages: return magnitude remains difficult to predict.	Plain linear regression would be simpler, but ElasticNet is more appropriate in a high-dimensional feature setting.
Large-move / event-day classification	A restricted classification setting focusing on days with stronger market movement or topic relevance.	It fits the idea that text signal may be stronger when the market context is more event-driven.	Advantages: more signal-focused and economically meaningful. Disadvantages: smaller sample size.	A full-sample-only approach would dilute potential signal with many low-information days.